

Tech Nation Visa - Exceptional Promise

Criteria: Mandatory Criteria (Innovation) & Optional Criteria 4 (Academic/Research)

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Evidence 2: Blind-Spot Decomposition Theory (BSDT) Framework

1. Research Innovation: The BSDT Mathematical Model

This document presents my proprietary mathematical framework, **Blind-Spot Decomposition Theory (BSDT)**, a domain-agnostic method for decomposing hidden model failure modes into four measurable channels. BSDT was developed alongside AMTTP as a separate research track rather than being embedded in the default production AML pipeline.

BSDT Four-Channel Decomposition:

$$\delta_C = 1 - \text{clip}\left(\frac{\|\mathbf{x} - \boldsymbol{\mu}_{\text{ref}}\|}{d_{\text{max}}}, 0, 1\right) \quad (\text{Camouflage})$$

$$\delta_G = \frac{\# \text{ near-zero features}}{p} \quad (\text{Feature Gap})$$

$$\delta_A = \sigma\left(\frac{d_{\text{Mahal}} - \tilde{d}_{\text{ref}}}{\tilde{d}_{\text{ref}}}\right) \quad (\text{Activity Anomaly})$$

$$\delta_T = \sigma\left(0.5 \cdot \left(\frac{d_{k\text{NN}}}{\tilde{d}_{k\text{NN},\text{ref}}} - 2\right)\right) \quad (\text{Temporal Novelty})$$

Blind-Spot Energy & Missed Fraud Likelihood Score:

$$E_{\text{BS}} = \sum_{i \in \{C, G, A, T\}} \delta_i^2 \quad (\text{Total blind-spot energy})$$

$$\text{MFLS} = \|\nabla E_{\text{BS}}\|_F \quad (\text{Gradient-norm score})$$

$$\text{Combined} = 0.5 \cdot \hat{E}_{\text{BS}} + 0.5 \cdot \widehat{\text{MFLS}} \quad (\text{Equal-weight average})$$

Figure 1: Mathematical formulation of the four-channel BSDT decomposition. Each channel captures an independent failure mode; the composite MFLS enables post-hoc diagnosis and optional correction without retraining the base model.

2. Code Implementation: From Theory to Practice

BSDT is not part of the default AMTTP transaction-scoring pipeline. I kept the production AML stack centered on the already high-performing distilled student models to avoid introducing unnecessary regulatory complexity into the core decision path. Instead, BSDT exists in the workspace as a separate research and enterprise analytics capability that customers can integrate into their own workflows where they need post-hoc failure analysis, model diagnostics, or adaptive feature layers.

Research Implementation (from `research/udl/udl/system_mode.py`):

```

class BSDTChannels:
    """Blind-Spot Detection Tensor --- four complementary channels."""
    def fit(self, X_ref: np.ndarray) -> 'BSDTChannels':
        self.mu_ = X_ref.mean(axis=0)
        self.d_max_ = max(float(np.linalg.norm(X_ref - self.mu_, axis=1).max()), self.eps)
        self.mahal_ref_median_ = float(np.median(self._mahalanobis(X_ref)))
        self.ref_knn_median_ = float(np.median(ref_dists[:, -1]))
    def mfls(self, X: np.ndarray) -> np.ndarray:
        return np.linalg.norm(self._gradient_vectors(X), axis=1)
    def score(self, X: np.ndarray) -> np.ndarray:
        e = self.energy(X)
        m = self.mfls(X)
        return 0.5 * (e / e.max()) + 0.5 * (m / m.max())

```

Figure 2: BSDT research implementation showing calibration on reference data, computation of blind-spot energy, and MFLS scoring inside the broader research workspace. This framework is maintained as an optional analytical layer rather than a default production scoring dependency.

3. Research Impact and Cross-Domain Relevance

BSDT should be understood as a complementary research framework, not as evidence that the production AMTTP models required boosting to perform well. The production student models are already sufficiently strong on their own. BSDT instead addresses a different problem: how to analyse hidden failure modes, construct optional post-hoc correction layers, and study instability across domains where customers or researchers may want more diagnostic depth than a regulator-facing default pipeline should carry.

The BSDT research programme has been validated extensively:

- In the 66-variant, 792-evaluation BSDT study, the best-performing family achieved **mean F1 = 0.787** while eliminating **88.1% of false positives**.
- In one flagship benchmark (Shuttle AMTTP-XGB93), the F2_QuadSurf variant reduced false alerts from 36,469 to 22 — a **99.94% false-positive reduction** — while lifting F1 from 0.134 to 0.968.
- In 10 of 12 dataset-model combinations, at least one BSDT variant achieved **zero missed fraud**, showing strong value as a second-stage alerting or diagnostic layer.

This research also connects directly to my broader “Geometry of System Collapse” work. In the wider workspace, the same analytical family is explored across financial stress, power-grid instability, and collapse dynamics, including ERCOT Texas grid experiments (AUC up to **0.9996**) and broader collapse case studies such as Terra/Luna-style dislocations. That cross-domain reach is important because it shows BSDT is not a narrow AML trick; it is part of a larger research programme on instability detection, adaptive friction, and system resilience.